

Fluid Dynamics

Homework 9

Complete coursework manuscript

Problem 1

Consider the ordinary differential equation

$$\varepsilon y'(x) + y = e^x, \quad x \geq 0, \quad \varepsilon \ll 1,$$

with initial value $y(0) = 0$.

- (a) Find the outer solution $y_o(x)$ to the equation corresponding to setting $\varepsilon = 0$, and verify that it does not satisfy the initial value provided.
- (b) Write down the problem for the inner solution $y_i(x)$ (i.e. the boundary layer.) What equation and which boundary conditions must it satisfy?
- (c) Solve the problem for y_i . Plot $y_i(x/\varepsilon)$ and $y_o(x)$ for $\varepsilon = 0.01$. Is there an interval where the two solutions nearly overlap?
- (d) Your problem for part (b) was a first order ODE with two boundary conditions, yet somehow you managed to solve it in part (c). Was it purely coincidental that this seemingly overdetermined problem had a solution? If not, why do you think it happened?

Solution to Problem 1:

- (a) **Outer solution.** Setting $\varepsilon = 0$ gives

$$y_o(x) = e^x,$$

which fails the initial condition since $y(0) = 0$ and $y_o(0) = 1 \neq 0$.

- (b) **The problem for inner solution.** Let $X = x/\varepsilon$ and $Y(X) = y(x)$. Then,

$$y'(x) = \frac{dy}{dx} = \frac{dy}{dX} \cdot \frac{dX}{dx} = \frac{1}{\varepsilon} Y'(X),$$

so the equation becomes

$$Y'(X) + Y(X) = e^{\varepsilon X}.$$

The inner boundary condition is $Y(0) = y(0) = 0$ and $Y(X) \rightarrow 1$ when $X \rightarrow \infty$.

- (c) **Solve and plot the problem.** Take $\varepsilon \rightarrow 0$, the equation is

$$Y'(X) + Y(X) = 1.$$

Solving this equation gives

$$Y(X) = 1 - Ce^{-X}$$

where C is a constant. Use initial condition $Y(0) = 0$,

$$0 = 1 - C \Rightarrow C = 1.$$

So,

$$Y(X) = 1 - e^{-X}.$$

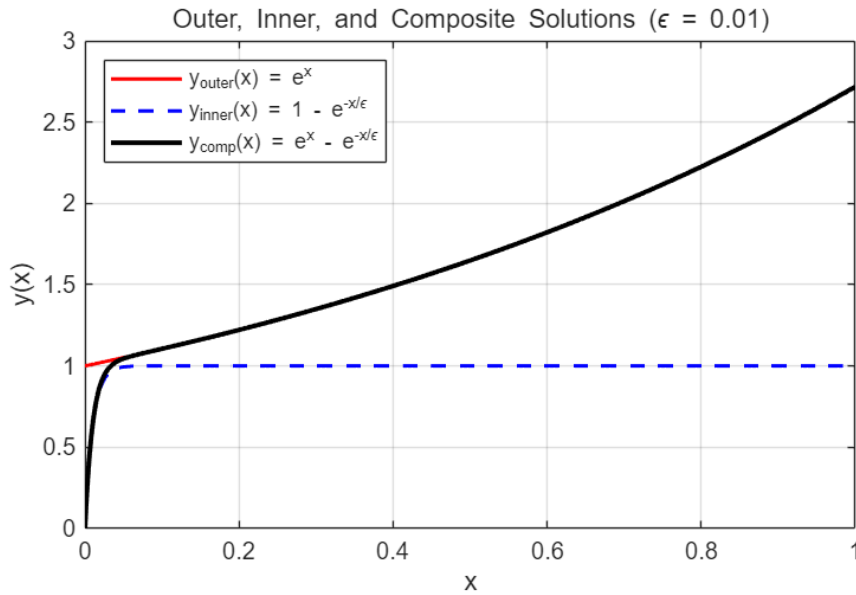
Returning to x gives

$$y_i(x) = 1 - e^{-x/\varepsilon}.$$

The composite solution is

$$y_{\text{composition}}(x) = y_i(x) + y_o(x) - y_{\text{overlap}}(x) = e^x + (1 - e^{-x/\varepsilon}) - 1 = e^x - e^{-x/\varepsilon},$$

where $y_{\text{overlap}} = 1$ because y_0 and y_i overlap at $y = 1$. Here is the plot for $\varepsilon = 0.01$: As for



the overlap region, when

$$x \ll 1$$

$$y_0 \rightarrow 1;$$

when

$$\varepsilon \ll x,$$

$$y_i = 1 - e^{-x/\varepsilon} \rightarrow 1.$$

Thus, the overlap interval is

$$\varepsilon \ll x \ll 1.$$

- (d) **Why the apparently overdetermined inner problem had a solution.** The inner condition $Y(0) = 0$ and the matching requirement $Y(\infty) = 1$ are not independent boundary conditions:

- The first reason is: one is the natural initial condition at $x = 0$, the other condition is an independent boundary condition which is a matching condition to the outer solution;
- The second reason is: it conforms to our general solution:

$$Y(X) = 1 - e^{-X},$$

when $X \rightarrow \infty, Y(X) \rightarrow 1$. In some sense, it can't be counted as an independent boundary condition.

Problem 2

Now consider the ODE

$$\varepsilon^2 y'' + x(y' - y^2) = 0,$$

for $x \in [-1, 1]$, with boundary conditions $y(-1) = -1$, $y(1) = 1$.

This appears quite complex, as it is second order, non-linear and has variable coefficients. Yet one can find an approximate solution valid for $\varepsilon \ll 1$.

Would your approximate solution have been any different if, instead of x , the factor of the last two terms had been $\tan(x)$?

Hints:

1. Set $\varepsilon = 0$ and solve the corresponding outer problem.
2. Verify that if you try to satisfy one of the two boundary conditions with your outer solution and then fit the other with a boundary layer, the signs conspire so that your inner solution is either a global constant or blows up as x moves into the outer domain.
3. Write two outer solutions, one satisfying the boundary condition on the left and the other on the right.
4. Make an educated guess on where the inner layer should be in order for the inner solution to decay to a constant in both directions away from it.
5. Find the right scaling for x in the inner layer so as to obtain a nontrivial balance of terms.
6. Solve the resulting equation for the inner solution and make it match the two outer ones.

Solution to Problem 2

1. **Step 1. Outer problem.** Set $\varepsilon = 0$. The ODE reduces to

$$x(y' - y^2) = 0.$$

For $x \neq 0$, we must have

$$y' = y^2, \\ \frac{dy}{y^2} = dx \quad \Rightarrow \quad -\frac{1}{y} = x + C.$$

Hence

$$y = -\frac{1}{x + C},$$

where C is a constant.

2. **Step 2. Left and right outer solutions.** The boundary conditions are

$$y(-1) = -1, \quad y(1) = 1.$$

We cannot satisfy both boundary conditions with one such expression, so we construct 2 boundary conditions y_L and y_R :

$$y_L(x) = -\frac{1}{x + C_L}, \quad \text{satisfying } y_L(-1) = -1, \\ y_R(x) = -\frac{1}{x + C_R}, \quad \text{satisfying } y_R(1) = 1.$$

From these, we get

$$y_L(-1) = -1 \implies -1 = -\frac{1}{-1 + C_L} \Rightarrow C_L = 2,$$

so $y_L = -\frac{1}{x+2}$.

Similarly,

$$y_R(1) = 1 \implies 1 = -\frac{1}{1 + C_R} \Rightarrow C_R = -2,$$

so $y_R = -\frac{1}{x-2}$.

3. Step 3. Location of the inner layer.

$$y_L(0) = -\frac{1}{2}, y_R(0) = \frac{1}{2}.$$

There is a jump at $x = 0$. Thus, we make an educated guess that the inner layer is

$$x = 0.$$

Because when the original equation equals 0, it is a point of degradation here.

4. Step 4. Rescale for x .

Define a stretched variable around $x = 0$:

$$x = \varepsilon X, \quad y(x) = Y(X).$$

Then

$$y' = \frac{1}{\varepsilon} Y', \quad y'' = \frac{1}{\varepsilon^2} Y''.$$

Substitute into the ODE:

$$\varepsilon^2 \left(\frac{1}{\varepsilon^2} Y'' \right) + (\varepsilon X) \left(\frac{1}{\varepsilon} Y' - Y^2 \right) = 0,$$

which simplifies to

$$Y'' + X(Y' - \varepsilon Y^2) = 0.$$

For $\varepsilon \ll 1$, the inner problem is

$$Y'' + XY' = 0.$$

5. Step 5. Solve the inner problem.

Let $Z = Y'$. Then, $Z' + XZ = 0 \Rightarrow Z = Ae^{-X^2/2}$ where A is a constant. Integrate Z gives:

$$Y(X) = A \int_0^X e^{-s^2/2} ds + B.$$

As $X \rightarrow -\infty$, $Y(X) \rightarrow y_L(0) = -1/2$, and as $X \rightarrow +\infty$, $Y(X) \rightarrow y_R(0) = 1/2$.

When $X \rightarrow \infty$,

$$I_\infty = \int_0^\infty e^{-s^2/2} ds = \sqrt{\frac{\pi}{2}}.$$

Thus,

$$Y(\infty) = AL_\infty + B, Y(-\infty) = -AL_\infty + B.$$

$$\Rightarrow 1/2 = AL_\infty + B, -1/2 = -AL_\infty + B.$$

$$\Rightarrow B = 0, A = 1/\sqrt{2\pi}$$

Thus,

$$y_{inner} = \frac{1}{\sqrt{2\pi}} \int_0^X e^{-s^2/2} ds = \frac{1}{2} \operatorname{erf}\left(\frac{x}{\sqrt{2}\varepsilon}\right).$$

Finally, the solution for this equation is

$$y(x) \sim \begin{cases} -\frac{1}{x+2}, & x \ll -\varepsilon, \\ \frac{1}{2} \operatorname{erf}\left(\frac{x}{\sqrt{2}\varepsilon}\right), & |x| = O(\varepsilon), \\ \frac{1}{2-x}, & x \gg \varepsilon. \end{cases}$$

6. Step 6. Effect of replacing x by $\tan x$.

If the coefficient x were replaced by $\tan x$, the structure would remain the same.

- For inner solution, when x is small, say $x = O(\varepsilon)$.

$$\tan(x) \sim x.$$

Thus the inner solution will not change!

- For outer solution, when $\tan(x) \neq 0$, which are the places

$$x \neq 0, \pm\pi, \dots$$

Within our interval $[-1, 1]$ the only zero is at $x = 0$, we must have

$$y' = y^2.$$

Thus, the outer solution will not change too!

Therefore, replacing x by $\tan x$ does not change the **solution**!